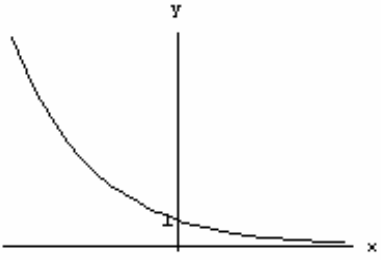
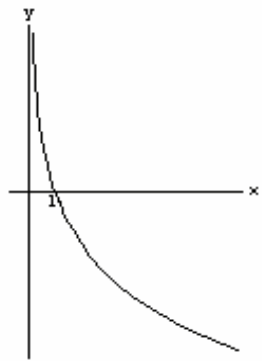
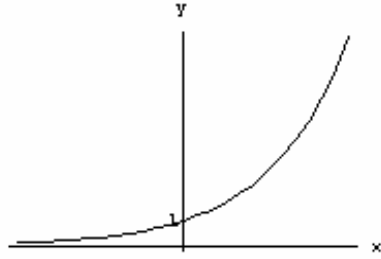
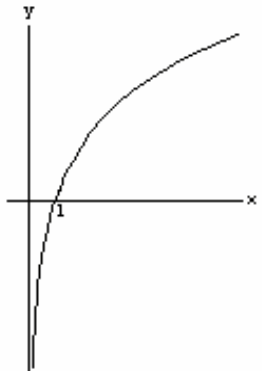
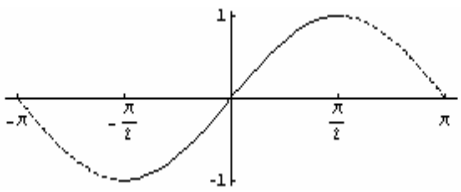
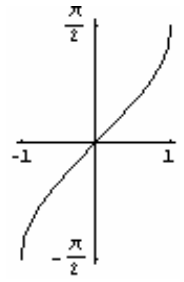
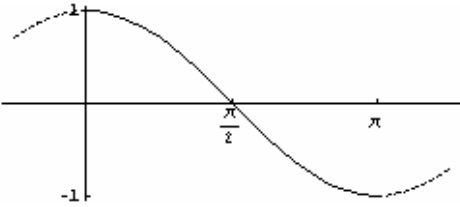
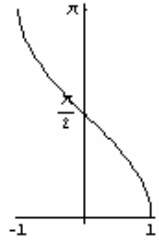
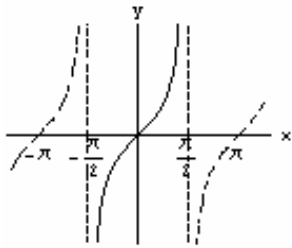
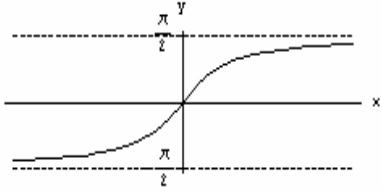
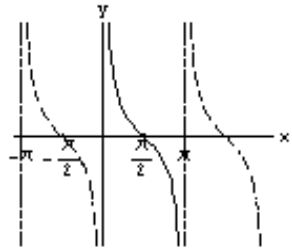
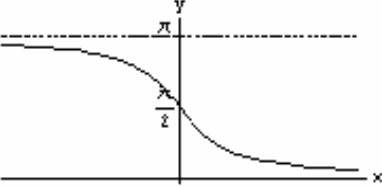


Funções Elementares

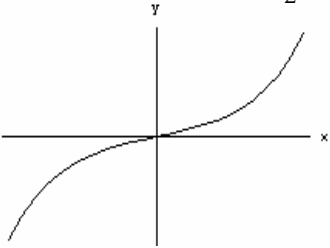
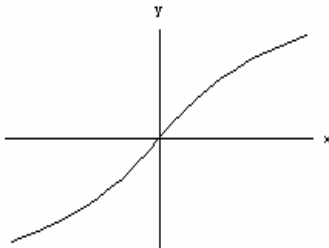
Função	Função Inversa
1. Função Exponencial $f : \mathbb{R} \rightarrow \mathbb{R}^+$ $x \mapsto a^x, a > 0 \wedge a \neq 1$	2. Função Logarítmica $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ $x \mapsto \log_a x, a > 0 \wedge a \neq 1$
$0 < a < 1$	
	
$a > 1$	
	

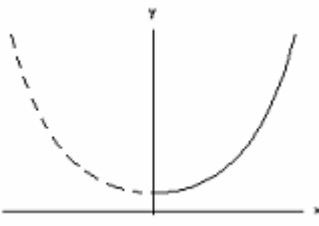
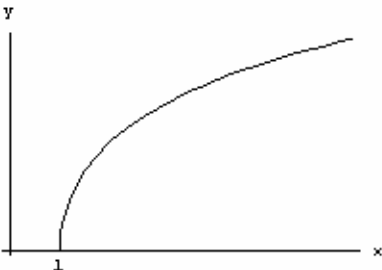
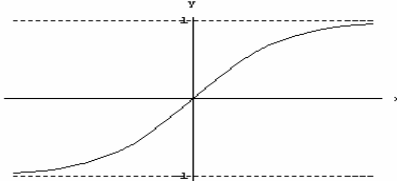
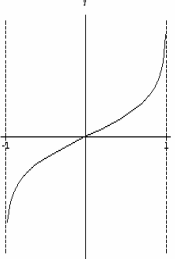
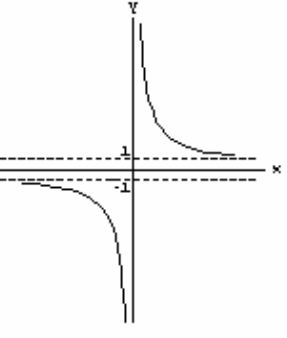
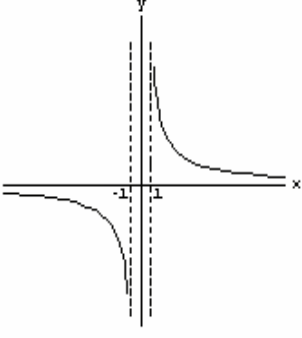
Funções Trigonômicas

3. Seno $f : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow [-1, 1]$ $x \mapsto \text{sen } x$	4. Arco Seno $f^{-1} : [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ $x \mapsto \arcsen x$
	

Função	Função Inversa
5. Cosseno $f : [0, \pi] \rightarrow [-1, 1]$ $x \mapsto \cos x$ 	6. Arco Cosseno $f^{-1} : [-1, 1] \rightarrow [0, \pi]$ $x \mapsto \arccos x$ 
7. Tangente $f :]-\frac{\pi}{2}, \frac{\pi}{2}[\rightarrow \mathbb{R}$ $x \mapsto \operatorname{tg} x$ 	8. Arco Tangente $f^{-1} : \mathbb{R} \rightarrow]-\frac{\pi}{2}, \frac{\pi}{2}[$ $x \mapsto \operatorname{arctg} x$ 
9. Cotangente $f :]0, \pi[\rightarrow \mathbb{R}$ $x \mapsto \operatorname{cotg} x$ 	10. Arco Cotangente $f^{-1} : \mathbb{R} \rightarrow]0, \pi[$ $x \mapsto \operatorname{arccotg} x$ 

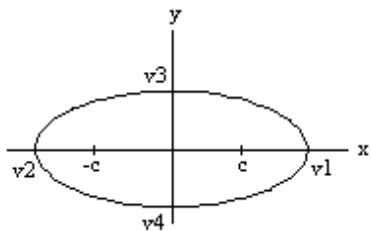
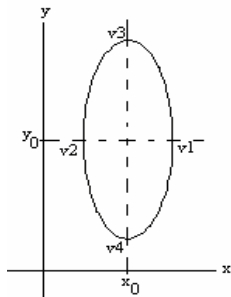
Funções Hiperbólicas

11. Seno Hiperbólico $f : \mathbb{R} \rightarrow \mathbb{R}$ $x \mapsto \operatorname{sh} x = \frac{e^x - e^{-x}}{2}$ 	12. Argumento Seno Hiperbólico $f^{-1} : \mathbb{R} \rightarrow \mathbb{R}$ $x \mapsto \operatorname{argsh} x$ 
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Função	Função Inversa
13. Cosseno Hiperbólico $f : \mathbb{R}_0^+ \rightarrow [1, +\infty[$ $x \mapsto \operatorname{ch} x = \frac{e^x + e^{-x}}{2}$ 	14. Argumento Cosseno Hiperbólico $f^{-1} : [1, +\infty[\rightarrow \mathbb{R}_0^+$ $x \mapsto \operatorname{argch} x$ 
15. Tangente Hiperbólica $f : \mathbb{R} \rightarrow]-1, 1[$ $x \mapsto \operatorname{th} x = \frac{\operatorname{sh} x}{\operatorname{ch} x} = \frac{e^{2x} - 1}{e^{2x} + 1}$ 	16. Argumento Tangente Hiperbólica $f^{-1} :]-1, 1[\rightarrow \mathbb{R}$ $x \mapsto \operatorname{argth} x$ 
17. Cotangente Hiperbólica $f : \mathbb{R} \setminus \{0\} \rightarrow]-\infty, -1[\cup]1, +\infty[$ $x \mapsto \operatorname{coth} x = \frac{\operatorname{ch} x}{\operatorname{sh} x} = \frac{e^{2x} + 1}{e^{2x} - 1}$ 	18. Argumento Cotangente Hiperbólica $f^{-1} :]-\infty, -1[\cup]1, +\infty[\rightarrow \mathbb{R} \setminus \{0\}$ $x \mapsto \operatorname{argcoth} x$ 

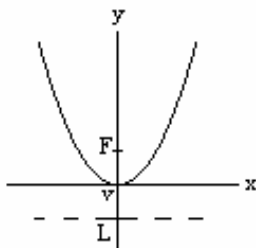
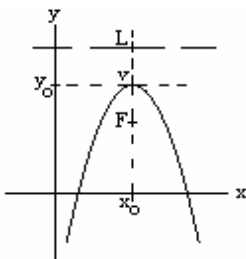
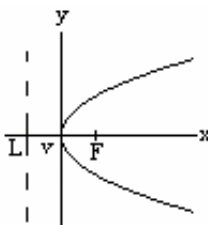
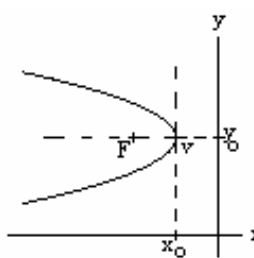
Cônicas

Elipse é o conjunto dos pontos do plano cuja soma das distâncias a dois pontos fixos (focos) é constante

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$  $2a$ = comprimento do eixo maior $2b$ = comprimento do eixo menor $2c$ = distância entre os focos $c^2 = a^2 - b^2$ Focos: $F_1 = (c, 0)$ e $F_2 = (-c, 0)$ Vértices: $v_1 = (a, 0)$, $v_2 = (-a, 0)$ $v_3 = (0, b)$, $v_4 = (0, -b)$ Excentricidade: $e = \frac{c}{a}$	$\frac{(x - x_0)^2}{a^2} + \frac{(y - y_0)^2}{b^2} = 1$  $2a$ = comprimento do eixo menor $2b$ = comprimento do eixo maior $2c$ = distância entre os focos $c^2 = b^2 - a^2$ Focos: $F_1 = (x_0, y_0 + c)$ e $F_2 = (x_0, y_0 - c)$ Vértices: $v_1 = (x_0 + a, y_0)$, $v_2 = (x_0 - a, y_0)$ $v_3 = (x_0, y_0 + b)$, $v_4 = (x_0, y_0 - b)$ Excentricidade: $e = \frac{c}{b}$
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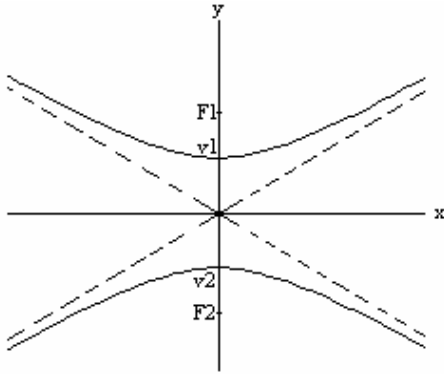
Parábola é o conjunto dos pontos do plano equidistantes de um ponto fixo F (foco) e de uma recta fixa L (directriz).

Excentricidade: $e = 1$; p = distância do foco à directriz

$x^2 = 2py$  Foco: $F = (0, p/2)$ Directriz: $y = -p/2$ Vértice: $v = (0, 0)$	$(x - x_0)^2 = -2p(y - y_0)$  Foco: $F = (x_0, y_0 - p/2)$ Directriz: $y = y_0 + p/2$ Vértice: $v = (x_0, y_0)$
$y^2 = 2px$  Foco: $F = (p/2, 0)$ Directriz: $x = -p/2$ Vértice: $v = (0, 0)$	$(y - y_0)^2 = -2p(x - x_0)$  Foco: $F = (x_0 - p/2, y_0)$ Directriz: $x = x_0 + p/2$ Vértice: $v = (x_0, y_0)$

Hipérbole é o conjunto dos pontos do plano tais que o módulo da diferença das distâncias a dois pontos fixos (focos) é constante e menor que a distância entre eles.

$$\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$$



$2a$ = comprimento do eixo não transverso

$2b$ = comprimento do eixo transverso

$2c$ = distância entre os focos

$$c^2 = a^2 + b^2$$

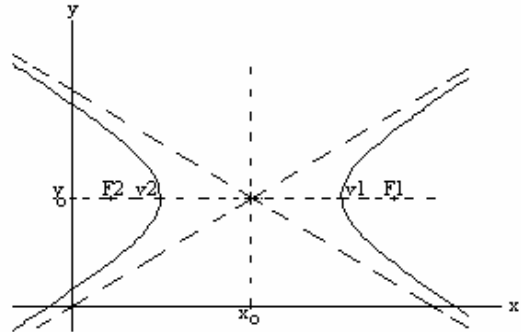
Focos: $F_1 = (0, c)$, $F_2 = (0, -c)$

Vértices: $v_1 = (0, b)$, $v_2 = (0, -b)$

Excentricidade: $e = \frac{c}{b}$

Assíntotas: $y = \pm \frac{b}{a}x$

$$\frac{(x - x_0)^2}{a^2} - \frac{(y - y_0)^2}{b^2} = 1$$



$2b$ = comprimento do eixo não transverso

$2a$ = comprimento do eixo transverso

$2c$ = distância entre os focos

$$c^2 = a^2 + b^2$$

Focos: $F_1 = (x_0 + c, y_0)$, $F_2 = (x_0 - c, y_0)$

Vértices: $v_1 = (x_0 + a, y_0)$, $v_2 = (x_0 - a, y_0)$

Excentricidade: $e = \frac{c}{b}$

Assíntotas: $y - y_0 = \pm \frac{b}{a}(x - x_0)$

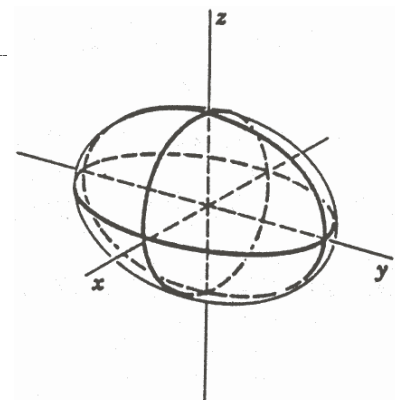
Superfícies Quádricas

Elipsóide $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

Intersecção com os planos coordenados:

$z = 0$	elipse	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
$x = 0$	elipse	$\frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$
$y = 0$	elipse	$\frac{x^2}{a^2} + \frac{z^2}{c^2} = 1$

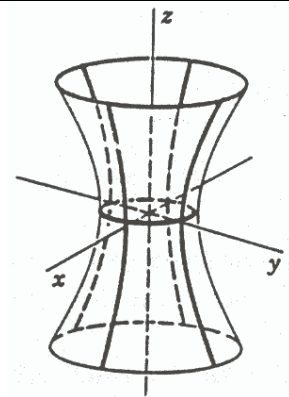
Obs. A **superfície esférica** de centro na origem é um caso particular de elipsóide com $a = b = c$



Hiperbolóide de uma folha $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$

Intersecção com os planos coordenados:

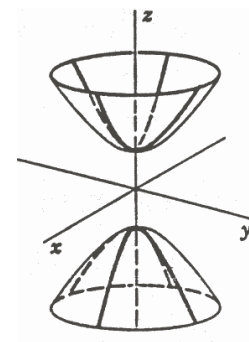
$z = 0$	elipse	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
$x = 0$	hipérbole	$\frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$
$y = 0$	hipérbole	$\frac{x^2}{a^2} - \frac{z^2}{c^2} = 1$



Hiperbolóide de duas folhas $\frac{z^2}{c^2} - \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Intersecção com os planos coordenados:

$z = 0$	a intersecção é vazia	
$x = 0$	hipérbole	$\frac{z^2}{c^2} - \frac{y^2}{b^2} = 1$
$y = 0$	hipérbole	$\frac{z^2}{c^2} - \frac{x^2}{a^2} = 1$



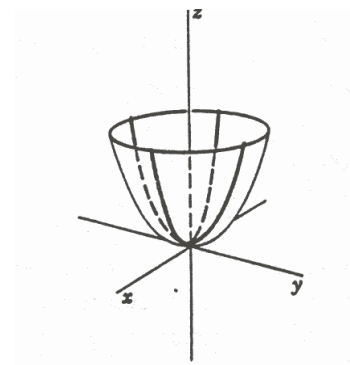
Parabolóide Elíptico $\frac{x^2}{a^2} + \frac{y^2}{b^2} - z = 0$

Intersecção com os planos coordenados:

$z = 0$	origem	$(0, 0, 0)$
$x = 0$	parábola	$z = \frac{y^2}{b^2}$
$y = 0$	parábola	$z = \frac{x^2}{a^2}$

Intersecção com um plano paralelo ao plano xoy

$z = z_0 > 0$	elipse	$z_0 = \frac{x^2}{a^2} + \frac{y^2}{b^2}$
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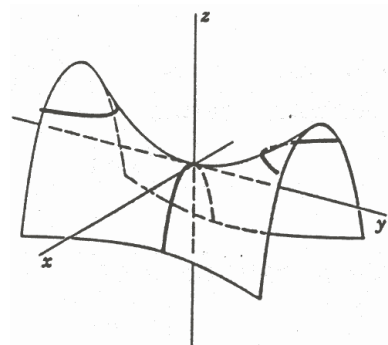
Parabolóide Hiperbólico	$\frac{y^2}{b^2} - \frac{x^2}{a^2} - z = 0$
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Intersecção com os planos coordenados:

$x = 0$	parábola	$z = \frac{y^2}{b^2}$
$y = 0$	parábola	$z = -\frac{x^2}{a^2}$
$z = 0$	rectas	$y = \frac{b}{a}x$ e $y = -\frac{b}{a}x$

Intersecção com um plano paralelo ao plano xoy :

$z = z_0$	hipérbole	$\frac{y^2}{b^2} - \frac{x^2}{a^2} = z_0$
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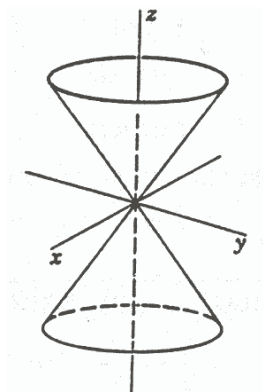
Cone Elíptico	$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$
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Intersecção com os planos coordenados:

$z = 0$	A intersecção é a origem
$x = 0$	rectas $z = \frac{c}{b}y$ e $z = -\frac{c}{b}y$
$y = 0$	rectas $z = \frac{c}{a}x$ e $z = -\frac{c}{a}x$

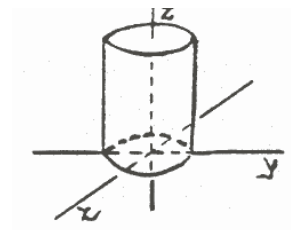
Intersecção com um plano paralelo ao plano xoy :

$z = z_0$	elipse	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z_0^2}{c^2}$
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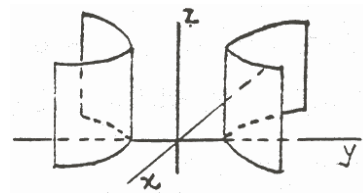
Cilindro Elíptico

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$



Cilindro Hiperbólico

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$



Cilindro Parabólico

$$y^2 = ax$$

